

Tutorial-8

Q1 $H = \frac{NI}{l} = \frac{100 \times 2}{2\pi \times 8} = \frac{100 \times 2}{2\pi \times 0.08}$

$= 397.88 \text{ At/m}$

For $H = 397.88 \text{ At/m}$

$B = 0.675$ from the B-H curve

$\phi = B \times A = 0.675 \times 0.009 = 6.075 \times 10^{-3} \text{ Wb}$

Q2 $\rightarrow$

$B = \frac{\phi}{A} = \frac{2 \times 10^{-4}}{1.3 \times 10^{-4}} = 1.5384$

(Assumed)

From the B-H curve, for $B_s = 1.5384$, $H_{ss} = 2400 \text{ At/m}$

From the magnetic circuit, applying Ampere's Circuital law,

$N_1 I_1 + N_2 I_2 = H_{ss} \cdot l_{ss} + H_g \cdot l_g$

or, $200 I_1 + 40 \times 0.3 = 2400 \times (2\pi \times 0.002) + \frac{\phi}{A \mu_0} \times 0.002$

or, $200 I_1 = 6967.4975 - 12$

$\therefore I_1 = 34.77 \text{ A}$

Q3 $\rightarrow B = \frac{\phi}{A} = \frac{12 \times 10^{-4}}{12 \times 10^{-4}} = 1 \text{ T}$

From the B-H curve, for $B = 1 \text{ T}$
 $H = 750 \text{ At/m}$ (Assumed)

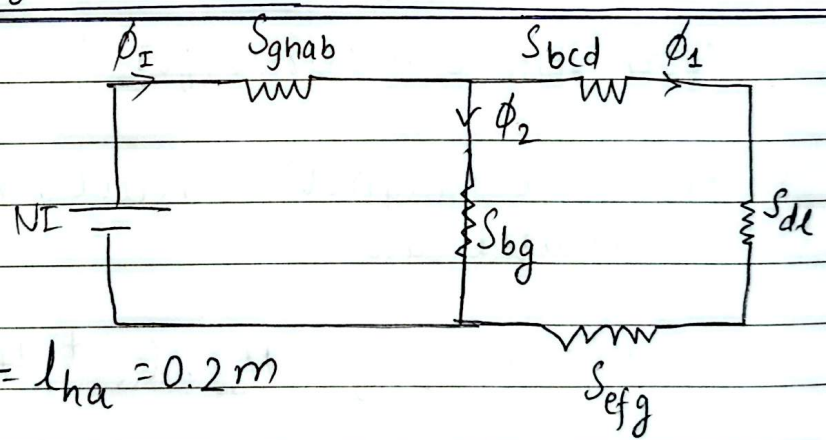
By Ampere's Circuital law,

$$N_1 I_1 + N_2 I_2 = H \cdot l$$

or, $2 \times N_1 + 30 \times 1 = 750 \times 0.2$
 $\therefore N_1 = 60$

$$\mu = \frac{B}{H} = \frac{1}{750} = 13.34 \times 10^{-4} \text{ Wb/Am}$$

Q4)



$$l_{ab} = l_{bg} = l_{gh} = l_{ha} = 0.2 \text{ m}$$

$$l_{bc} = l_{fg} = 0.1 \text{ m}$$

$$l_{cd} = l_{ef} = 0.099 \text{ m}$$

$$l_{de} = 0.002 \text{ m} = l_g$$

For the airgap, $\phi_g = 2 \times 10^{-4} \text{ Wb}$

$$\therefore \phi_1 = \phi_g = 2 \times 10^{-4} \text{ Wb}$$

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$$\begin{aligned}\text{Flux density } (B_r) \text{ at the right limb} &= \frac{\phi_1}{A} \\ &= \frac{2 \times 10^{-4}}{5 \times 10^{-4}} \\ &= 0.4 \text{ T}\end{aligned}$$

From the BH curve, For $B_r = 0.4 \text{ T}$
 $H_r = 65 \text{ At/m}$

For the air gap,

$$B_g = \frac{\phi_1}{0.4} = 0.4 \text{ T}$$

$$\begin{aligned}H_g &= \frac{B_g}{\mu_0} = \frac{0.4}{4\pi \times 10^{-7}} \\ &= 318309.8862 \cdot \text{At/m}\end{aligned}$$

$\therefore$ MMF for the right limb including the airgap is

$$\begin{aligned}F_R &= H_g l_g + H_r (l_{bc} + l_{gf} + l_{cd} + l_{ef}) \\ &= (318309.8862 \times 0.002) + (65 \times 0.398) \\ &= 636.6197 + 25.87 \\ &= 662.4897 \text{ At}\end{aligned}$$

Since the right limb is parallel to central limb, the mmf of the central limb,

$$\begin{aligned}F_c &= F_R = 662.4897 \text{ At} \\ H_c l_{cg} &= 662.4897 \text{ At}\end{aligned}$$

$$\therefore H_c = \frac{662.4897}{0.2} = 3313.0985 \text{ At/m}$$

For $H_c = 3313.0985 \text{ At/m}$,

$B_c = 1.55 \text{ T}$ (Assumed from B-H curve)

$$\therefore \phi_2 = B_c \cdot A_c = 1.55 \times 2 \times 10^{-4} \\ = 3.1 \times 10^{-4} \text{ Wb}$$

For parallel magnetic circuit,

$$\therefore \phi_T = \phi_1 + \phi_2 = (2 \times 10^{-4}) + (3.1 \times 10^{-4}) \\ = 5.1 \times 10^{-4} \text{ Wb}$$

The flux density (B_L) for the left limb is

$$B_L = \frac{\phi_T}{A_L} = \frac{5.1 \times 10^{-4}}{5 \times 10^{-4}} = 1.02 \text{ T}$$

From the B-H curve, for $B_L = 1.02 \text{ T}$, $H = 190 \text{ At/m}$
The mmf for the left limb,

$$F_L = H_L \times (l_{ab} + l_{ab} + l_{ng}) = 190 \times 0.6 = 114 \text{ At}$$

Now,

Considering loop 1 and applying
Ampere's Circuital law,

$$NI = F_L + F_c$$

$$\text{or, } I = \frac{F_L + F_c}{N} = \frac{114 + 662.4897}{200} \\ = 3.88 \text{ A}$$

Q5)

Soln:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\therefore \frac{N_2}{N_1} = \frac{120}{2400} = 0.05 = K \text{ (transformation ratio)}$$

$$\text{Secondary current} = \frac{10 \text{ KVA}}{120} = 83.33 \text{ A}$$

$$\text{Primary current} = \frac{10 \text{ KVA}}{2400} = 4.167 \text{ A}$$

Q6,

Soln:

$$\frac{E_s}{E_p} = \frac{N_s}{N_p} \Rightarrow E_s = \frac{N_s}{N_p} \times E_p = \frac{30}{240} \times 25 = 3.125 \text{ V}$$

$$\phi_m = \frac{E_p}{4.44 f N_p} = \frac{25}{4.44 \times 60 \times 240} = 3.91 \times 10^{-4} \text{ Wb}$$

For,

$$\phi_m = 3.91 \times 10^{-4} \text{ Wb}$$

$$f = \frac{25}{4.44 \times 3.91 \times 10^{-4} \times 240} = 58.3 \text{ Hz}$$